Assignment 2

Feature Engineering using Spark, Hyperparameter Tuning using Ray Tune, and Experiment Tracking using MLflow

Objective

The objective of this assignment is to build an end-to-end machine learning pipeline using Spark, Ray Tune, and MLflow. You will:

- Perform feature engineering using Spark ML.
- Save the processed dataset in Parquet format.
- Train a binary classification model.
- Perform hyperparameter tuning using Ray Tune.
- Track experiments using MLflow.
- Register the best-performing model in the MLflow Model Registry.

Dataset

The path to access **Bank** dataset (`train.csv and test.csv`) on server is
{"/storage/data/datasets/assignment_2_dataset/"}.
You can also access the dataset from
{https://www.kaggle.com/competitions/playground-series-s5e8/overview}.
Target variable is `y`.

Part A: Feature Engineering using Spark (20 Marks)

Implement a Python script named `features_<roll_no>.py`.

The script should:

1. Read train.csv and test.csv using Spark.
2. Implement a preprocessing pipeline
   (a) Handle missing values.
   (b) Encode categorical columns using StringIndexer
   (c) Assemble features using VectorAssembler.
   (d) Scale numerical features when appropriate for the selected model.
3. Fit the preprocessing pipeline on the training dataset and apply the fitted pipeline to both the training and test datasets.
4. Create the target label for the training dataset. The processed test dataset should contain only the transformed features.
5. Save the processed train and test dataset as two separate Parquet files (train_processed_<roll_no>.parquet, test_processed_<roll_no>.parquet).

Part B: Model Training using Ray Tune (30 Marks)

Implement a Python script named `train_<roll_no>.py`.

The script should:

1. Load the processed training Parquet dataset.
2. Split the processed training data into training and validation sets (80:20)
3. Build a binary classification model. You can choose any ML/DL model.
4. Tune the model using Ray Tune by maximizing the validation F1-score.
5. Evaluate the best-performing model on the validation set.
6. Register the best model in MLflow.

Perform at least **10 hyperparameter tuning trials**.

Part B: MLflow Logging (20 Marks)

Create an experiment named `assignment_2_<roll_no>`. Each Ray Tune trial must be logged as a separate MLflow run. Parameters should be logged.

Metrics

- Training loss and validation loss if using pytorch model
- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC
- Training time

Artifacts

- Trained model
- Confusion matrix
- Classification report
- Predictions on validation set

Part C: Model Registration (20 Marks)

- Register the model from the Ray Tune trial with the highest validation F1-score in the MLflow Model Registry using the name Assignment2Classifier_<roll_no>. `Assignment2Classifier_<roll_no>`.
- Demonstrate loading the registered model and performing inference on the processed test.csv dataset. Save the predictions as a CSV file (predictions_<roll_no>.csv).

Part D: Report (10 Marks)

Prepare a report (2-3 pages) summarizing:

- Spark preprocessing pipeline
- Hyperparameter search space
- Best hyperparameters
- Final model performance
- Include the generated predictions_<roll_no>.csv file and briefly describe the inference process.
- Screenshots of MLflow experiments and Model Registry

Files to Submit

- features_<roll_no>.py
- train_<roll_no>.py
- report_<roll_no>.pdf
- predictions_<roll_no>.csv